Errors-in-variables regression for mixed Euclidean and non-Euclidean predictors

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1 The problem

Most errors-in-variables regression methods are developed for Euclidean predictors. This paper studies nonparametric regression when both Euclidean and non-Euclidean predictors are contaminated by measurement error. The predictor takes values in

$$\mathbb{R}^d \times \mathbb{G}.$$

2 The setup

Let

$$X = (X_1, X_2) \in \mathbb{R}^d \times \mathbb{G},$$

where $X_1 \in \mathbb{R}^d$ is a Euclidean predictor and $X_2 \in \mathbb{G}$ is a non-Euclidean predictor. The regression model is

$$Y = m(X) + \epsilon, \qquad E(\epsilon \mid X) = 0.$$

The latent predictor X is not observed directly. Instead, we observe

$$Z = (Z_1, Z_2) = (U_1 + X_1, U_2 \circ X_2),$$

where $U_1 \in \mathbb{R}^d$ and $U_2 \in \mathbb{G}$ are unobservable measurement errors. The assumptions include

$$U \perp (X, \epsilon), \qquad U_1 \perp U_2,$$

and the measurement error densities f_{U_1} and f_{U_2} are known.

Given i.i.d. observations

$$\{(Y_i, Z_i) : 1 \le i \le n\}, \qquad Z_i = (Z_{i,1}, Z_{i,2}) \in \mathbb{R}^d \times \mathbb{G},$$

the goal is to estimate

$$m(x) = E(Y \mid X = x), \qquad x = (x_1, x_2) \in \mathbb{R}^d \times \mathbb{G}.$$

3 The main idea

The estimator combines Euclidean and non-Euclidean deconvolution.

For the Euclidean component,

$$Z_1 = X_1 + U_1$$

leads to the deconvolution kernel

$$K_h^D(s) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle s,t\rangle} \frac{\phi_K(t)}{\phi_{U_1}(t/h)} \, dt.$$

For the non-Euclidean component, Fourier analysis is carried out on the compact Lie group $\mathbb{G}$. For each irreducible representation $\sigma \in \widehat{\mathbb{G}}$, the model

$$Z_2 = U_2 \circ X_2$$

allows the effect of U_2 to be corrected by multiplication with

$$\phi_{U_2}(\sigma^M)^{-1}.$$

This gives the truncated deconvolution kernel

$$K_{T_n}(x_2, z_2) = \sum_{\sigma \in \widehat{\mathbb{G}}: k_\sigma < T_n} d_\sigma \operatorname{Tr}\left\{\sigma^M(z_2^{-1})\phi_{U_2}(\sigma^M)^{-1}\sigma^M(x_2)\right\},$$

where k_σ is the Casimir spectrum and T_n is a smoothing parameter.

The regression estimator is

$$\widehat{m}(x) = \widehat{f}_X(x)^{-1} \frac{1}{nh^d} \sum_{i=1}^{n} K_h^D\left(\frac{x_1 - Z_{i,1}}{h}\right) \operatorname{Re}\left\{K_{T_n}(x_2, Z_{i,2})\right\} Y_i,$$

where

$$\widehat{f}_X(x) = \frac{1}{nh^d} \sum_{i=1}^{n} K_h^D\left(\frac{x_1 - Z_{i,1}}{h}\right) \operatorname{Re}\left\{K_{T_n}(x_2, Z_{i,2})\right\}.$$

The denominator estimates $f_X(x)$, while the numerator estimates $m(x)f_X(x)$. Their ratio therefore estimates $m(x)$.

The deconvolution kernels satisfy

$$E\left[K_h^D\left(\frac{x_1 - Z_1}{h}\right) K_{T_n}(x_2, Z_2) \mid X\right] = K\left(\frac{x_1 - X_1}{h}\right) K_{T_n}^*(x_2, X_2).$$

This property corrects the bias caused by measurement error by making the kernel score behave, in conditional expectation, as if X were observed.

4 The main results

The paper derives pointwise, uniform, and L^2 convergence rates for $\widehat{m}$. It also establishes the asymptotic distribution of $\widehat{m}(x)$ under the ordinary-smooth condition, which can be used to construct asymptotic confidence intervals. The paper also discusses the case in which the measurement error distributions are unknown. Simulation studies compare the proposed estimator with a naive estimator that ignores measurement error.

5 My takeaways

- Since $\widehat{m}(x)$ divides by $\widehat{f}_X(x)$, what lower-bound condition on $f_X(x)$ is needed to keep the estimator stable and to establish uniform convergence?

References

[1] J. M. Jeon. Errors-in-variables regression for mixed Euclidean and non-Euclidean predictors. *Journal of Nonparametric Statistics*, 37(2):282–318, 2025. https://doi.org/10.1080/10485252.2024.2378897.